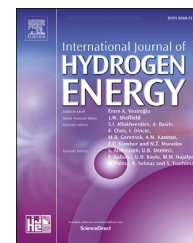


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Construction and operation of hydrogen energy utilization system for a zero emission building

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ABSTRACT

A bench-scale stationary hydrogen energy utilization system with renewable energy (RE) that realizes a zero emission building (ZEB) is presented. To facilitate compactness, safety, and mild operation conditions, a polymer electrolyte membrane (PEM) electrolyzer for hydrogen production (5 Nm³/h), PEM fuel cells (FC) for hydrogen use (3.5 kW), and metal hydride (MH) tanks for hydrogen storage (80 Nm³) are incorporated. Each hydrogen apparatus and Li-ion batteries (20 kW/20 kWh) are installed in a 12-ft. container and 20-kW photovoltaic panels provide power. A building energy management system (BEMS) controlled these system components in an integrated manner. The PEM Ely and FC have fast start-up and high efficiency under partial load operations, indicating suitability for daily start-stop operations. An AB-type TiFe-based alloy (520 kg) is used as the MH (not an AB₅-type rare earth alloy that has been commonly used in bench-scale hydrogen store) because, in addition to being low-cost, it is non-hazardous material under Japanese regulations. The results of a 24-h operation experiment verify ZEB attainment. PEM FC and TiFe-based tanks thermal integration results indicate that hydrogen use operation is achievable without external heat sources.

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Introduction

Greenhouse gas emissions from carbon dioxide (CO₂), methane (CH₄), and other substances are largely a result of fossil fuel use, and contribute greatly to global warming and climate change. In Japan, almost 40% of the total emitted CO₂ is due to energy consumption within buildings, which greatly exceeds the contributions from mobility sources (~16%) [1]. In December 2017, the Japanese government released a “Basic Hydrogen Strategy” [2], which provides a roadmap for the

realization of a hydrogen-based society (this roadmap was revised March 12, 2019 [3]). Included in the roadmap are zero-emission buildings (ZEBs), which have an effective annual energy consumption of zero through energy conservation and renewable energy (RE) power generation [4]. Accordingly, reducing CO₂ emissions from buildings is an absolute necessity.

In many cases, stationary hydrogen energy utilization systems with RE such as photovoltaic (PV) panels and wind turbines consist of an electrolyzer (Ely) for hydrogen

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production, as well as hydrogen storage tanks and fuel cells (FC) for hydrogen use [5–11]. Batteries for short-term energy storage are often included in the stationary system [12]. To achieve the ZEB, we have developed a bench-scale hydrogen energy system with RE [13]; simulation results for the system design and its operations are detailed in our other work [14]. In that system, 20-kW PV panels are used for RE power generation because they can be easily installed near or on buildings. Our hydrogen system consists of Ely (5 Nm³/h), metal hydride (MH) tanks (80 Nm³), FC (3.5 kW), and batteries (20 kW/20 kWh). This bench-scale system was aimed to achieve a ZEB for a small low-rise building (total floor area ~1000 m²; demand: ~5 kW) [14].

To use hydrogen in urban environments (i.e., strict regulation areas), our stationary system prioritizes compactness, safety, and mild operation conditions (e.g. < 1 MPa and 70 °C). We thus selected MH tanks that can store hydrogen compactly and operate near ambient conditions [15–18]. The MH in the tanks is not classified as a hazardous material under the Fire Service Act of Japan. We suggest that MH hydrogen storage (MHHS) is the least regulated storage method and thus suitable for use in urban areas. The maximum hydrogen pressure is controlled such that it is less than 1 MPa (Gauge), to adhere to the High Pressure Gas Safety Regulation of Japan [19].

AB₅-type rare-earth alloys (e.g., LaNi₅ and Mm(NiCoAl)₅) are the most typical hydrogen storage alloys and have been used as bench-scale MHHS for stationary systems [5–11,20–22], because they possess easy activation, small hysteresis, and rapid kinetics. However, AB₅ alloys include expensive rare-earth metals and become pulverized after a few ab/desorption cycles [23]. The pulverized alloy grains are very small (~1 μm) and active, and are classified as hazardous materials in Japan. Therefore, MHHS units using the hazardous material must be placed at a safe distance (several meters) from the building, meaning that the required space is often too large for use in urban building areas [24].

We focused on the AB-type TiFe-based alloy [25] because of the low cost of the materials and abundance of resources. Moreover, many of the AB-type alloys have high pulverization resistance and thus are not classified as hazardous materials. The disadvantages of the alloys include large hysteresis, second plateau, and difficult activation treatments [22,23,25,26]. However, we investigated the activation processes of a bench-scale MH tank filled with the TiFe-based alloy near ambient conditions [27], and succeeded in developing the alloy with a small second plateau and easy activation. One of the unique features of our hydrogen energy utilization system is that the alloy grains (total: 520 kg) were placed in each tank (TiFe-based tank).

The second feature of our system is thermal integration between the FC and TiFe-based tanks. Owing to the endothermic hydrogen desorption of the alloy, it is necessary to heat the alloy bed for continuous hydrogen release. As the FC simultaneously generates power and heat, desorption of the TiFe-based tanks can be operated by using the FC heat output. Many thermal integrations of FC and MH tanks have been studied, and the MH tanks were commonly filled with AB₅ alloys (AB₅ tanks) [28–33]. However, few studies have investigated integrating FC and TiFe-based tanks. In particular, no studies to our knowledge have examined bench-scale

integrated systems, of which our integrated system would be the largest scale (3.5 kW and 520 kg).

The third feature of our hydrogen system is that it can be operated automatically by a building energy management system (BEMS) that was developed by Shimizu Co. called “Smart BEMS” [34]. Smart BEMS plans energy management based on weather and demand forecasts and minimizes grid power consumption (⇔ the CO₂ emissions). Based on the excess PV power and the power demand of the building, the hydrogen apparatuses and charge-discharge of the batteries can be controlled by the Smart BEMS.

In this paper, a bench-scale hydrogen energy utilization system was constructed to realize a ZEB. Our system was aimed to be employed in a small-rise building to experimentally realize a ZEB. We describe the specifications of each system component and detail the technical aspects that save cost/space in the system design. Subsequently, we present experimental results from 24-h operation in fine weather conditions. Details on the thermal integration of the FC and TiFe-based tanks are also presented. We believe our results will help enable practical applications of the stationary hydrogen system.

Hydrogen energy utilization system for a ZEB

System appearance and specifications

Fig. 1 shows a photograph of the system. Each system component was installed in a 12-ft. container to facilitate easy relocation into any building areas. On the roofs are 3.75-kW PV panels (Panasonic, HIT™) to provide power to the auxiliaries. The auxiliaries' power is 3.3 kW and 0.5 kW under hydrogen production (Ely + MH) and utilization (MH + FC) operations, respectively. To suppress temperature increases inside the container, ventilating fans are placed in each container; to conserve power, air conditioners are not employed. All hydrogen pipelines consist of 3/8 or 1/2-inch stainless steel tubes. To release hydrogen safely outside in an emergency



Fig. 1 – Photograph of the hydrogen energy utilization system for a ZEB. In front of the battery container 20-kW PV panels are located (not shown).

situation, bent pipe lines are also placed on each container's roof, except for the battery container.

Table 1 summarizes the system component specifications. In front of the battery container 20-kW PV panels (Panasonic, HIT™) are placed. For short-term power storage/supply and emergency start-up power supply, Li-ion rechargeable batteries (CAPTEX, BS-0100_11P8S_12S) containing high power and voltage are employed. Each hydrogen apparatus is described in detail in the ensuing sections.

Hydrogen production-electrolyzer

For the system to be employed in urban areas, compactness is a high priority. Hydrogen is produced by excess PV power, so the apparatus must be suitable for daily start/stop (DSS) and partial load (PL) operations. Thus, we selected a compact polymer electrolyte membrane (PEM) Ely (Hitachi Zosen, HYDROSPRING™) and pure water, instead of an alkaline Ely that occupies a larger space and has a poisonous solution. The PEM Ely is capable of hydrogen production of 5 Nm³/h with DC power of 26 kW. Maximum hydrogen supply pressure to the TiFe-based tanks is 0.97 MPaG to conform with the Japan High Pressure Gas Safety Act [19].

Fig. 2 shows a photograph of the PEM Ely installed in the container. The cell stacks consist of 26 electrochemical cells. Other specifications include a radiator (Mitsubishi Electric, EWF-50FTA) used for the cell stacks cooling to save space and cost, and no cooling tower, allowing for easy relocation. Two dehumidifiers (Japan Pionics, JHP-10E-S) are used for dehumidification of hydrogen from the PEM Ely. The dew point of the hydrogen supplied to the TiFe-based tanks is less than −50 °C. To reduce hydrogen emissions outside the container, a gas tank stores hydrogen at a high dew point (see H₂ tank in Fig. 2), using the under-regeneration operation of a dehumidifier. The air valve compressor is the same as that used in the TiFe-based tanks, to save space and cost. Further, in this system, electrolysis waste heat is used for heat storage and heating the MH tanks.

Fig. 3 shows the start-up of the PEM Ely. The run-up operation begins about 2 min after the start, followed by rated operation (5 Nm³/h) approximately 5 min after the start. The Ely has a fast start-up, which indicates that it is suitable for the DSS operation. Furthermore, the fast start-up is not dependent on the cell stacks' temperature.

The hydrogen conversion efficiency of the PEM Ely at the cell stacks' temperature of 65–70 °C is presented in Fig. 4. The

efficiency includes the auxiliaries' power, such as the water circulation pump, radiator, ventilating fan, and the compressor. Around 80% or more is obtainable over 150 A and the conversion efficiency has a maximum value of 87% at 250 A. These results show that hydrogen production with high efficiency could happen with fluctuating PV power except at low current values (<100 A). The efficiency decreased about 10% after a decrease of 30 °C.

Hydrogen use - Fuel cells

Compactness is also a high priority for the hydrogen use apparatus. This apparatus is operated in accordance with building demand, meaning it must be suitable for DSS and PL operations. For this purpose, we selected a water-cooled PEM FC (Toshiba, H₂Rex™).

Fig. 5 shows a photograph of the PEM FC installed in the container. The radiators (Toshiba Carrier, VP-304SNX) are used to cool the cell stacks, which makes it possible to build a small water-cooled close loop. The integration lines between the PEM FC and TiFe-based tanks are shown in Fig. 6. The heat exchange between the PEM FC and heat transfer liquid (60% brine) is conducted at the plate heat exchanger (ALFA LAVAL, AN14-20H), which is indicated in Fig. 6(a); subsequently, the heat is supplied to the TiFe-based tanks. The integration operation is fully described in Section Thermal integration of PEM FC - Metal hydride tanks.

Fig. 7 shows the start-up of the PEM FC. The run-up operation (1.25 kW) starts approximately 7 min after startup, and the rated-operation (3.5 kW) approximately 10 min after startup. This result demonstrates that the PEM FC has a fast start-up and the operation is not dependent on the cell stacks' temperature. As seen in Fig. 7, the power responsiveness from the lowest/maximum to maximum/lowest output occurs in tens of seconds. The electrical efficiency at each power output is 55%, calculated based on a lower heating value (LHV). These results show that the PEM FC is suitable for the DSS and PL operations.

Hydrogen storage - Metal hydride tanks

As with the Ely and FC, compactness is a high priority for the hydrogen storage apparatus. To comply with Japanese regulations, no use of hazardous material and high pressure gas (>1 MPaG) are allowed. Therefore, we selected the MHHS, which has advantages such as mild operation conditions, reliability, low power consumption, and ultra-high purity.

Table 1 – Specifications of each system component.

Functions	System component	Specifications
H ₂ Production	PEM electrolyzer	26 kW ⇔ ≥5 Nm ³ /h (500A)
H ₂ Storage	TiFe-based alloy (No rare earth metals)	H ₂ amount: 80 Nm ³ for 2 units Absorption: 5 Nm ³ /h (20 °C) Desorption: 3 Nm ³ /h (60 °C)
H ₂ Use	PEM Fuel cell (Pure H ₂ type)	Power output : 3.5 kW Electrical efficiency(LHV) : 55 %
Power storage	Li ion Battery	Capacity :10 kWh × 2 Power output: 10 kW × 2
RE power	Amorphous Si + Single crystal Si	10 kW × 2, 0.25 kW × 15 (Containers roofs)

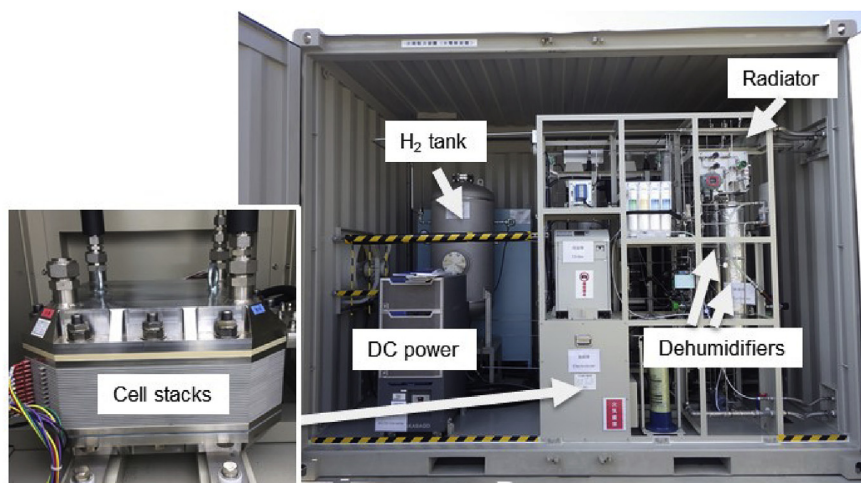


Fig. 2 – Photograph of the hydrogen production apparatus (PEM Ely) installed in a 12 ft container. The left photo shows the cell stacks (DC power of 26 kW [500 A] ⇔ 5 Nm³/h). The radiator and the dehumidifiers are pictured at right, while the hydrogen gas tank (H₂ tank) shown in the middle have a capacity of 330 L.

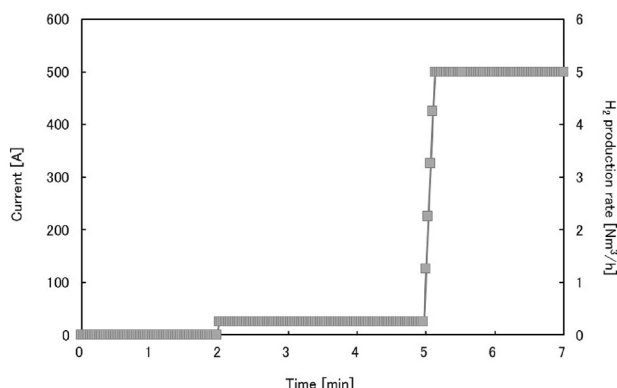


Fig. 3 – Graph of current and hydrogen flow rate at start-up of the PEM Ely.

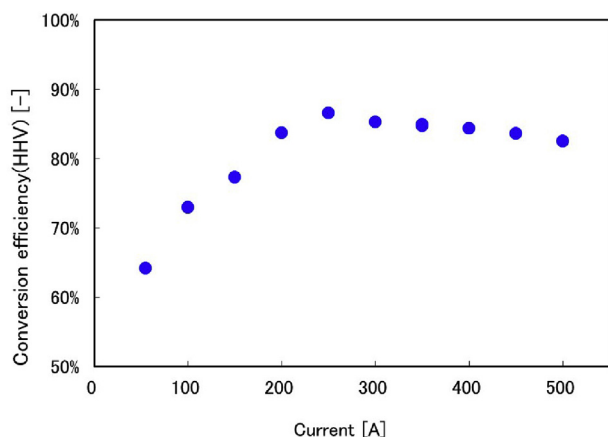


Fig. 4 – Hydrogen conversion efficiency of the PEM Ely at the cell stacks' temperature of 65–70 °C. Each efficiency is calculated based on a higher heating value (HHV).

Each MH tank is filled with the TiFe-based alloy TiFeMn (Japan Metals & Chemicals), with a few Mn substituting for

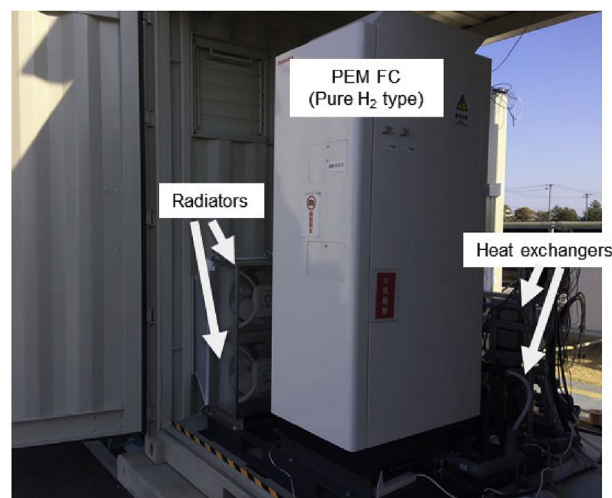


Fig. 5 – Photograph of the hydrogen use apparatus (PEM FC) installed in a 12 ft container. The FC has a maximum power output of 3.5 kW (⇔ 2.1 Nm³/h), hydrogen supply pressure of <50 kPa, and FC outlet water temperature of 60 °C. Two radiators and heat exchangers are placed as pictured.

another transition metal that was prepared using a high-frequency melting furnace under an Ar atmosphere. The as-cast alloy was equilibrated through annealing for 24 h at 1100 °C. The developed alloy (~5 g) is able to absorb hydrogen immediately after vacuuming for 1 h at 70 °C. Its maximum hydrogen storage capacity, as described by the hydrogen to metal atom ratio (H/M), is 0.9. The effective hydrogen storage amount is 1.4 wt% ($0.1 \leq H/M \leq 0.83$) under absorption at 30 °C and desorption at 45 °C, which is larger than that of AB5-type alloys (~1 wt%) [35]. The ignition properties of the alloy grains were evaluated after several ab/desorption cycles using a

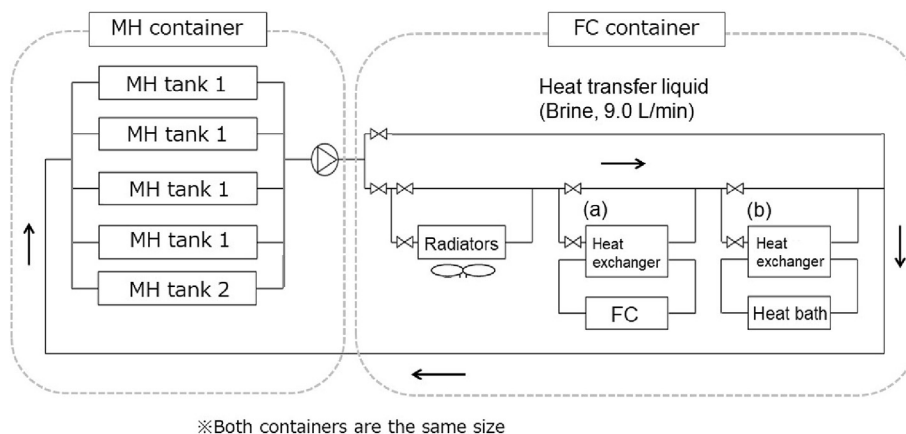


Fig. 6 – Schematic of the thermal integration lines between the PEM FC and TiFe-based tanks. Each MH tank (1 and 2) is filled with TiFe-based alloy of 65 and 260 kg, respectively. The heat transfer liquid is 60% brine (9.0 L/min). Radiators are employed to cool the PEM FC and the MH tanks. The heat exchangers in (a) and (b) are the same model. Note that both MH and FC containers are the same size.

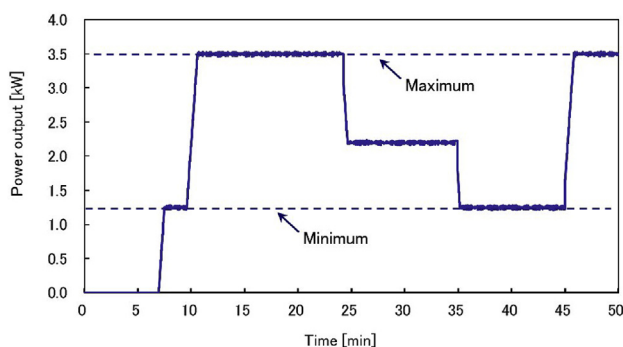


Fig. 7 – Graph of the start-up and power output responsiveness of the PEM FC.



Fig. 8 – Photograph of the hydrogen storage apparatus (MH tanks) installed in a 12 ft container. There are two types of tanks (1 and 2). Tank 1 is filled with 65 kg of the TiFe-based alloy using four canisters, while Tank 2 is filled with 260 kg in one canister. The total hydrogen storage capacity is 80 Nm³.

small gas flame ignition test [36], and the alloy grains were not ignited. For this reason, the alloy was not classified as a hazardous material under the Fire Service Act of Japan.

The TiFe-based tanks installed in the container are shown in Fig. 8. Two types of MH tanks are used and the hydrogen storage capacity of each is 40 Nm³. MH tank 1 is one unit with four canisters to enable high hydrogen flow rate operation. MH tank 2 is used for low flow rate operations and was developed to save cost and space. Both MH tanks were water heated/cooled and had a cylindrical shape. The activation processes were carried out at less than 70 °C and 0.97 MPaG; the details have been reported in Endo et al. [27]. The radiators for cooling the alloy beds were the same as those used in the PEM FC (see Fig. 6).

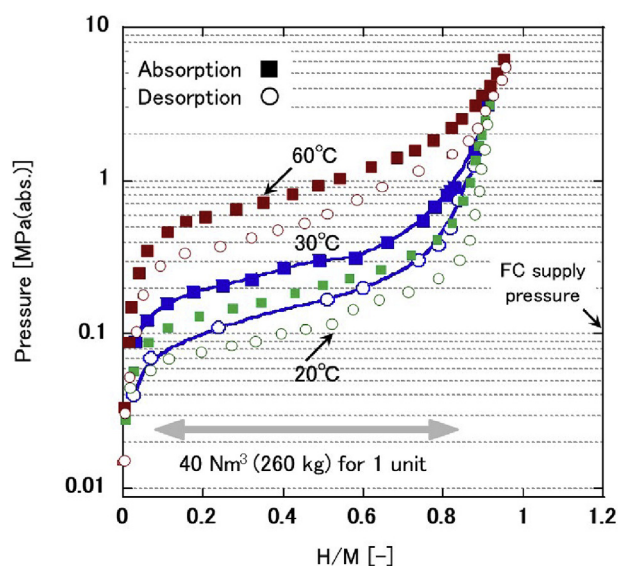


Fig. 9 – PCI properties of the TiFe-based alloy contained in each tank at 20, 30, and 60 °C. Filled square (■) and open circle (○) symbols correspond to absorption and desorption, respectively. The effective hydrogen storage amount of 40 Nm³ for one unit (260 kg) is shown by the double arrow. The FC supply pressure is indicated by an arrow on the right-hand vertical axis.

Fig. 9 shows pressure-composition isotherm (PCI) properties of the TiFe-based alloy used to fill the tanks. As mentioned above, many TiFe-based alloys have large hysteresis and a second plateau, which are generally disadvantages [22,23,25,26]. Although the hysteresis remained, the second plateau was smaller than a binary alloy [25] (see Fig. 9). The hydrogen supply pressure from the Ely to the TiFe-based tanks is 0.97 MPa. Meanwhile, the hydrogen pressure of the FC operation has to be less than 50 kPaG (minimum pressure of 15 kPaG); as such, the hydrogen supply pressure from the TiFe-based tanks to the FC was lowered by the pressure reducer. Therefore, we only had to set the hydrogen pressure of the TiFe-based tanks below 0.97 MPaG and over 15 kPaG under the hydrogen production (Ely + MH) and utilization (MH + FC) operations, respectively. Overall, the hysteresis and second plateau of the developed alloy are not major disadvantages; in reality, the TiFe-based tanks were able to be operated similarly to the AB5 tanks. This is a very important point for MHHS using the TiFe-based tanks.

Results and discussion

24-h operation in fine weather conditions

A 24-h operation was carried out automatically by the Smart BEMS [34]. Each system component was controlled by the Smart BEMS to maximally reduce grid power consumption. Based on a weather forecast from the previous day, the Smart BEMS planned the operation schedule for every 30 min up to 48 h ahead, then modified the operation schedule based on updated weather forecasts at 7:00, 12:00, and 19:00 on the operating day. The Smart BEMS was able to maintain real-time control within several seconds; that is, the Smart BEMS specified when the batteries charged and discharged for supply-demand balance based on PV power and building demand. The details of the control logics have been reported in our other work [14].

Fig. 10 shows the energy flow of the hydrogen system with PV. The direct-use PV power has the highest priority. In

addition, when excess PV power is generated, the batteries operate for short-term/small quantity energy storage. Finally, power-to-hydrogen was selected for long-term/large quantity energy storage. The FC power supply was designed for long-term operations under night or cloudy/rain conditions. In this experiment, we consider the power consumption of the control room and the auxiliaries' power as building demand (~5 kW), which is close to that of our target low-rise building.

Fig. 11 shows the 24-h operation results under fine weather conditions. The left vertical axis indicates the power supply (upper) and the demand/energy storage (lower). The gray line shows the energy storage rate (hydrogen + battery) on the right vertical axis. Although the operation was controlled every 30 min, the time variations are shown every 1 h because there was almost no difference in the results between the two time intervals. The building demand (blue) was supplied by the FC power (dark blue) from 0:00 to 7:00. Because the demand was smaller than the PV power output, excess PV power (green) was generated after 8:00 and the batteries charged the excess PV power at 8:00 (orange). Subsequently, hydrogen production (light blue) occurred from 9:00 to 16:00, during which period the charge-discharge operations of the Li-ion batteries were due to the real-time control. The energy storage rate (gray line) depended mainly on the hydrogen storage amount because the difference between the charge and discharge of the batteries was small. On account of little grid power (red), the planned operation was successfully carried out; that is, ZEB was achieved.

The overall system efficiency p was calculated using the following equation:

$$p = \frac{\text{Demand} + \Delta E_{(+)} }{PV \text{ power} - \Delta E_{(-)}} \times 100 \quad (1)$$

where PV power is the total PV power output [126.9 (kWh/24 h)], Demand is the total building demand [79.4 (kWh/24 h)], and ΔE is the energy storage difference [kWh/24 h]. In this experiment, ΔE was -5.5 kWh; as a result, $p = 59.9$, i.e., 60% was obtainable, which agreed with our simulation results [14]. p does not include building heat usage; this result indicates that the heat usage is important for further high efficiency,

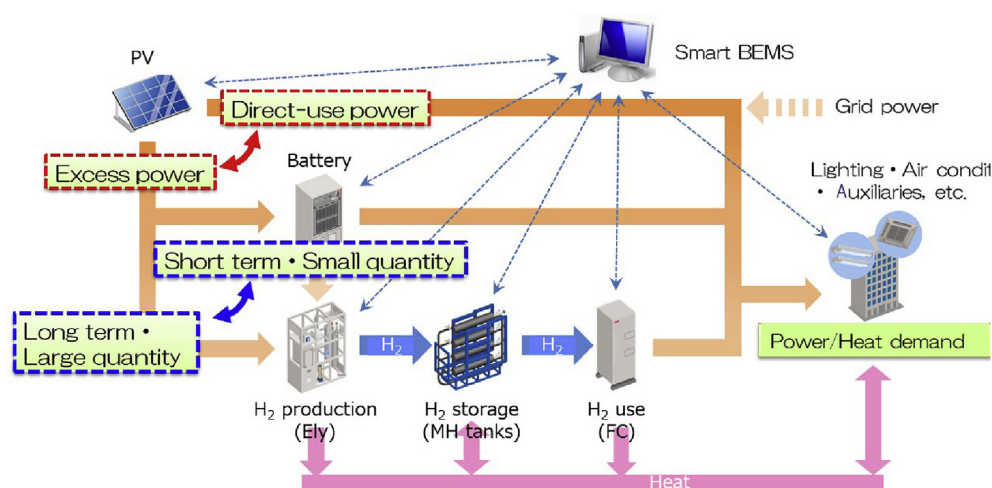


Fig. 10 – Schematic of the energy flow of the hydrogen energy utilization system with PV.

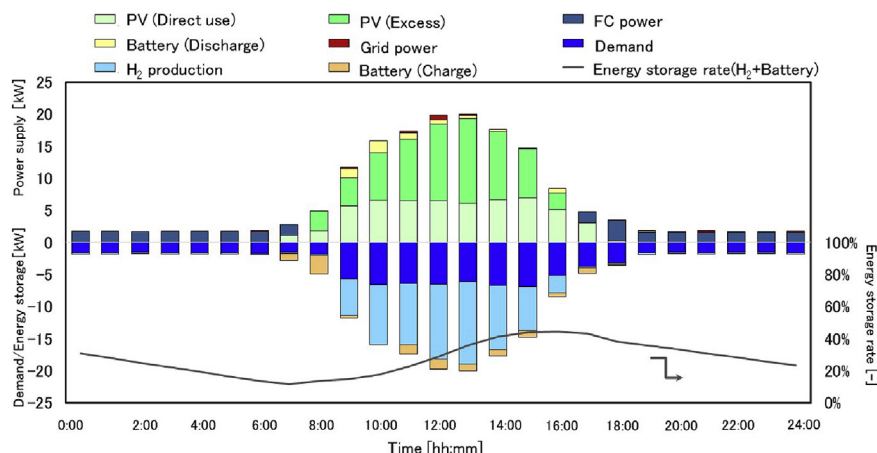


Fig. 11 – Power supply and demand/energy storage under 24 h operation with fine weather condition. The gray line is the energy storage rate (hydrogen + battery).

and the hydrogen energy utilization system is suitable for buildings with high heat demand (e.g. hotels and hospitals). Although only the 24-h operation during fine day is shown here, we confirmed that operations in other weather conditions produced similarly successful results [37].

Thermal integration of PEM FC - Metal hydride tanks

The thermal integration between the PEM FC and TiFe-based tanks was demonstrated. Integrated systems have been reported [28–33], with many using AB₅-type alloys and others using AB₂-type alloys [33,38–40]. However, few studies have used TiFe-based tanks. To the best of our knowledge, our integration scale is the largest with 3.5 kW power and a 520 kg alloy.

MH Tank 1 was used in this experiment (see Figs. 6 and 8); the results are shown in Fig. 12. The initial temperature, pressure, and hydrogen storage of the tanks were 30 °C, 0.15 MPa, and 20 Nm³ (H/M = 0.5), respectively. Here, the temperature of the tanks refers to that of the alloy bed. The FC power output and hydrogen flow rate were 3.5 kW and

2.1 Nm³/h, respectively. The FC outlet water temperature increased up to 60 °C in 20 min after the power output started, and then held steady at 60 °C. The heat exchange between FC and the heat transfer liquid was carried out at the heat exchanger indicated in Fig. 6(a), after which the heat was supplied to the alloy tanks. The heat supply temperature to the tanks was constant at approximately 42 °C when the FC outlet water temperature was 60 °C. The flow rate of the heat transfer liquid was 9.0 L/min.

As shown in Fig. 12, the temperature of the tanks increased at 20 min after the FC power output, and pressure increased accordingly. The temperature increased to 42 °C after 3 h of operation, and temperature and pressure subsequently decreased gradually. Although the hydrogen flow rate (2.1 Nm³/h) and heat supply temperature (42 °C) were constant during the operation, the tank temperature decreased. This is because the thermal conductivity of the alloy bed decreased with decreasing hydrogen storage amount; the alloy grains contracted as the alloy desorbed hydrogen, and the void fraction of the alloy bed increased [27,41]. For this reason, the thermal conductivity decreased and the heat exchange between the alloy bed and the heat transfer liquid also decreased. Despite the decreasing temperature and pressure of the TiFe-based tanks, full desorption was carried out using the FC heat output (pink line in Fig. 12).

The heat exchange, q , between the heat transfer liquid and the FC circulation water was calculated using the following equation:

$$q = C_p \cdot \rho \cdot \Delta T \cdot v \quad (2)$$

where C_p is the specific heat of the heat transfer liquid at constant pressure [3.85 kJ/(kg·K)], ρ is density (1.02 kg/m³), ΔT is the temperature difference between the inlet and outlet heat transfer liquid of the heat exchanger, and v is the flow rate of the circulation brine (9.0 L/min). q was almost constant at 2.4 kW during the FC power output of 3.5 kW, and the heat recovery efficiency was 40% (LHV). The endothermic reaction heat of the TiFe-based alloy was 0.91 kW, estimated from the van't Hoff plot in Fig. 9. The FC heat output was more than twice the reaction heat. Each FC heat output at power outputs

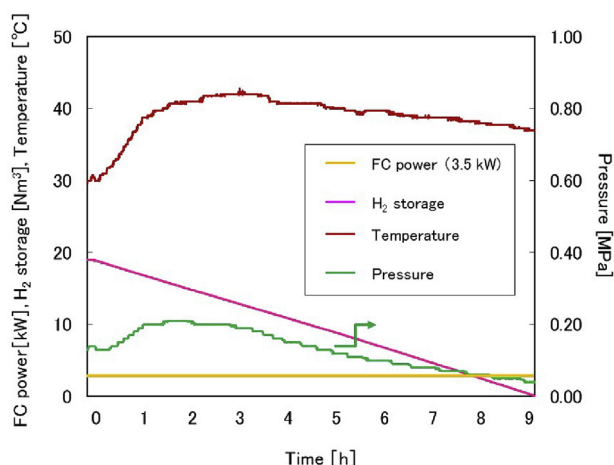


Fig. 12 – Results for thermal integration operation of the PEM FC-TiFe-based tanks.

of 1.25–3.5 kW was sufficiently higher than each reaction heat. Accordingly, this result indicates that a hydrogen use operation (MH + FC) can be carried out without external heat sources.

Conclusions

We constructed a bench-scale hydrogen energy utilization system with RE for a ZEB, with the aim of employing it in a small low-rise building. Because of practical considerations in urban areas, the priorities of our hydrogen system were compactness, safety, and mild operation conditions. Consequently, PEM Ely and FC, MH tanks, and Li-ion batteries were selected. Each system component was installed in a 12 ft container that could be moved easily into any building area, and was controlled automatically by the BEMS. Both PEM Ely and FC exhibited fast start-up and high efficiency under partial load operations. We selected a TiFe-based alloy for MH hydrogen storage because of metal cost and Japanese regulations. The tanks (TiFe-based tanks) were filled with 520 kg of alloy grains. A 24-h operation on a fine day was carried out by the BEMS, and results showed that there was nearly zero grid power consumption, i.e. ZEB was realized. The thermal integration of PEM FC and TiFe-based tanks showed that full desorption could be completed through partial use of FC heat output. Our results are relevant to the practical application of stationary hydrogen energy systems in urban building areas.

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Abbreviations

CO ₂	carbon dioxide
CH ₄	methane
ZEB	zero energy buildings
RE	renewable energy
PV	photovoltaic
Ely	electrolyzer
FC	fuel cell
MH	metal hydride
MHHS	metal hydride hydrogen storage
MPaG	MPa (Gauge)
TiFe-based tank	tank filled the TiFe-based alloy
AB ₅ tank	tank filled the AB ₅ -type alloy
BEMS	building energy management system
DSS	daily start stop
PL	partial load
PEM	polymer electrolyte membrane
HHV	higher heating value
LHV	lower heating value

H/M	hydrogen to metal atom ratio
PCI	pressure-composition-isotherm

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